What Drives Life Expectancy?

A Regression Analysis of WHO Country-Level Health Data

Course: Data Analysis I

Lecturer: Dr. HAS Sothea

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Team Members

KHOV Seangngy (2026571) — Role
PHAUT Sovannvatanak (2026525) — Role

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1 Introduction

1.1 Overview

Life expectancy at birth is one of the most widely used indicators of a country's overall health and development. It reflects not only the quality of healthcare, but also education, income, nutrition, and disease burden. Understanding which of these factors matter most helps governments and health organizations decide where to invest limited resources. This project analyzes the Life Expectancy (WHO) dataset from Kaggle, which combines health and socioeconomic indicators for about 190 countries from 2000 to 2015. The data contains both quantitative variables (such as adult mortality, immunization coverage, GDP, and schooling) and qualitative variables (such as country and development status). It also contains missing and invalid values, so careful cleaning is required before modeling. Using a data analysis pipeline of preprocessing, exploratory analysis, regression modeling, and evaluation, we identify the strongest drivers of life expectancy and assess how accurately it can be predicted.

1.2 Objective

The primary objective of this project is to build and evaluate regression models that predict a country's life expectancy from health, economic and social indicators, and to identify which factors have the strongest influence.

The specific objectives are:

1. Clean and preprocess the data by handling missing values, invalid values, outliers and inconsistent column names.
2. Explore the relationships between life expectancy and the other variables using statistical summaries, visualizations and correlation analysis.
3. Develop and compare several regression models against a baseline using RMSE, MAE and R^2.
4. Interpret the best model to determine the main drivers of life expectancy and discuss their practical implications.

1.3 Dataset Summary

Table 1: Dataset overview.

Item	Description
Source	Kaggle – Life Expectancy (WHO) (Kumar Rajarshi), compiled from WHO and United Nations data. `https://www.kaggle.com/datasets/kumarajarshi/life-expectancy-who`
Coverage	193 countries, 2000–2015 (one row per country per year)
Size	2938 rows × 22 columns before cleaning → 2928 rows × XX columns after cleaning (including engineered features)
Target	`life_expectancy` (years)
Quantitative	19 variables, e.g. `adult_mortality`, `bmi`, `gdp`, `schooling`
Qualitative	`country`, `status` (Developed/Developing); `year` is ordinal; engineered: `region`, `period`, `income_tier`
Data quality issues	Missing values in 14 columns, implausible values (e.g. `bmi`, `measles`), inconsistent column names
Task	Supervised regression

2 Data Preprocessing

2.1 Initial Inspection

The dataset was loaded with `pandas` from `Life Expectancy Data.csv`. It contains **2938 rows and 22 columns** covering 193 countries from 2000 to 2015. Key observations from the inspection:

- **Column names:** several names contained trailing or repeated spaces (for example `"Life expectancy "`), so all names were stripped and converted to snake_case.
- **Data types:** 20 columns are numeric (`float64`/`int64`) and 2 are categorical (`object`): `country` and `status`.
- **Target variable:** `life_expectancy` has a mean of 69.225 years, a standard deviation of 9.524, and ranges from 36.3 to 89.
- **Suspicious values:** the summary statistics show implausible values in several columns: `bmi` ranges from 1 to 87.3 (unrealistic for a national average), `adult_mortality` has a minimum of 1, `measles` reaches 212,183 although it is reported per 1000 population, `percentage_expenditure` reaches 19,479 although it is a percentage, `population` has a minimum of 34, and `income_composition_of_resources` and `schooling` have minimum values of 0. These are handled in Section 2.4.

- **Missing values:** 14 of the 22 columns contain missing values. The highest rates are `population` (22.19%), `hepatitis_b` (18.82%) and `gdp` (15.25%), examined in the next subsection. No duplicate rows were found.

```
data = pd.read_csv("Life Expectancy Data.csv")
print(data.shape)

data.columns = (data.columns.str.strip().str.lower()
                .str.replace(r"[\s\-]+", "_", regex=True))
data.info()
data.describe().T
```

Table 2: Summary statistics of all numeric features (before cleaning).

Feature	Count	Mean	Std	Min	Median	Max
year	2938	2007.52	4.61	2000	2008	2015
life_expectancy	2928	69.22	9.52	36.3	72.1	89.0
adult_mortality	2928	164.80	124.29	1	144	723
infant_deaths	2938	30.30	117.93	0	3	1800
alcohol	2744	4.60	4.05	0.01	3.76	17.87
percentage_expenditure	2938	738.25	1987.92	0	64.91	19479.91
hepatitis_b	2385	80.94	25.07	1	92	99
measles	2938	2419.59	11467.27	0	17	212183
bmi	2904	38.32	20.04	1	43.5	87.3
under_five_deaths	2938	42.04	160.45	0	4	2500
polio	2919	82.55	23.43	3	93	99
total_expenditure	2712	5.94	2.50	0.37	5.76	17.6
diphtheria	2919	82.32	23.72	2	93	99
hiv_aids	2938	1.74	5.08	0.1	0.1	50.6
gdp	2490	7483.16	14270.17	1.68	1766.95	119172.7
population	2286	1.28×10^7	6.10×10^7	34	1.39×10^6	1.29×10^9
thinness_1_19_years	2904	4.84	4.42	0.1	3.3	27.7
thinness_5_9_years	2904	4.87	4.51	0.1	3.3	28.6
income_composition_of_resources	2771	0.628	0.211	0	0.677	0.948
schooling	2775	11.99	3.36	0	12.3	20.7

2.2 Duplicates

A check for duplicate rows was performed using `duplicated()`. The result confirmed that **no duplicate rows** exist in the dataset (0 duplicates found). Since each row should represent one country in one year, we also checked that the `(country, year)` pairs are unique. Therefore, no rows were removed.

```
print(data.duplicated().sum())                          # 0
print(data.duplicated(subset=["country", "year"]).sum())    # expected 0
```

2.3 Missing Values

Fourteen of the 22 columns contain missing values. The most affected columns are `population` (22.19%), `hepatitis_b` (18.82%) and `gdp` (15.25%), while the target `life_expectancy` is missing in only 10 rows (0.34%).

Table 3: Missing values per column ($n = 2938$).

Column	Missing (n)	Missing (%)
population	652	22.19
hepatitis_b	553	18.82
gdp	448	15.25
total_expenditure	226	7.69
alcohol	194	6.60
income_composition_of_resources	167	5.68
schooling	163	5.55
bmi	34	1.16
thinness_1_19_years	34	1.16
thinness_5_9_years	34	1.16
polio	19	0.65
diphtheria	19	0.65
life_expectancy	10	0.34
adult_mortality	10	0.34

Figure: bar chart of missing percentage by column

Pattern of missingness. Missing values are not spread evenly: XX (describe what you found, e.g. missing values concentrate in a few countries or years, or appear together in the same rows). Because the probability of a value being missing depends on observed variables such as `country` and `status`, the data is treated as Missing At Random (MAR) rather than Missing Completely At Random.

Handling strategy.

- **Target variable:** the 10 rows with missing `life_expectancy` were dropped, since imputing the target would bias the model.
- **Predictors:** missing values were imputed with the median of the same `country`, which preserves country-level differences. If a country had no observed value at all, the median of its `status` group (Developed or Developing) was used instead.
- **Why the median:** most of these variables are right-skewed (e.g. `gdp`, `population`), so the median is more robust than the mean.

```
data = data.dropna(subset=["life_expectancy"])

num_cols = data.select_dtypes("number").columns.drop("life_expectancy")
for col in num_cols:
    data[col] = data.groupby("country")[col].transform(lambda s: s.fillna(s.median())
        )
    data[col] = data.groupby("status")[col].transform(lambda s: s.fillna(s.median()))

print(data.isnull().sum().sum())    # 0
```

After imputation, no missing values remained in the dataset.

2.4 Invalid Data

List each impossible value, the rule used to flag it, how many rows, and the fix.

Table 4: Invalid-value rules (examples to check).

Column	Rule	Action
bmi	outside plausible national mean range	set to NaN, impute
infant_deaths	0 with large population	flag, impute
schooling / income_composition	equals 0	treat as missing

2.5 Outliers

IQR method on skewed variables; state which were kept (genuine) and which were capped, with reasons.

Figure: boxplots before vs. after treatment

2.6 Feature Engineering and Encoding

Engineered categorical columns (region, period, income tier), binary encoding of `status`, one-hot encoding of the rest, log transform of skewed features.

3 Exploratory Data Analysis (EDA)

3.1 Univariate Analysis

Target distribution, then key features. One-line takeaway under each figure.

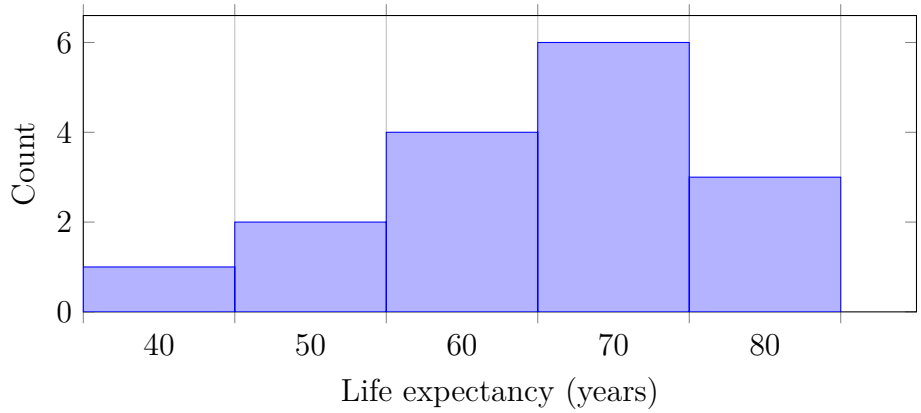


Figure 1: Illustrative histogram only: replace with your real plot (or embed `plt.savefig` PNG).

3.2 Bivariate Analysis

Numeric vs. target (scatter + trend), categorical vs. target (boxplots by `status`, `region`), trend over `year`.

Figure: scatter plots of top features vs. life expectancy

3.3 Correlation Analysis

Pearson vs. Spearman, multicollinearity pairs, chi-square / ANOVA for categorical variables.

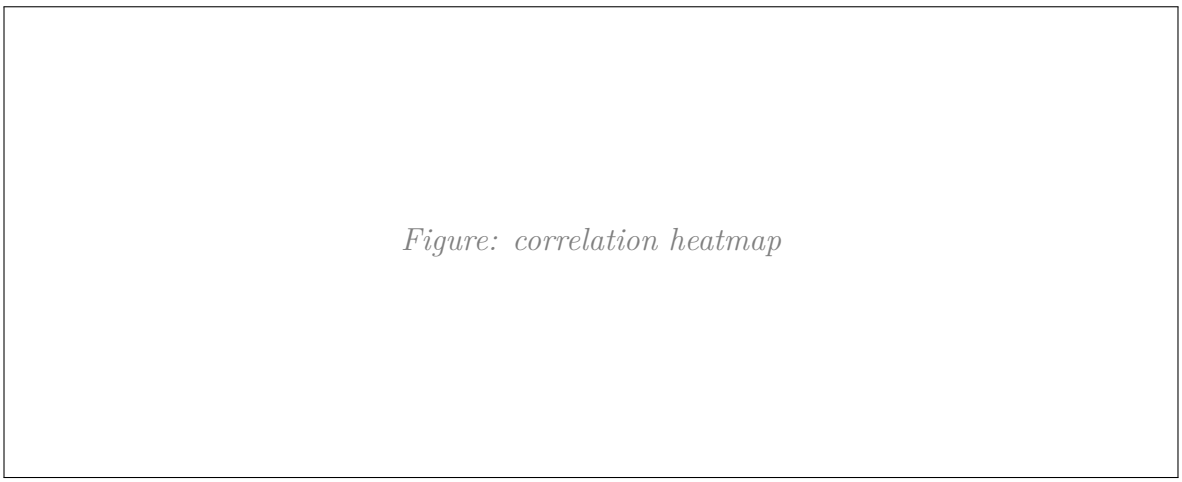


Table 5: Strongest correlations with the target (fill in).

Feature	Pearson	Spearman
schooling	XX	XX
adult_mortality	XX	XX
…		

3.4 EDA Hypotheses

- H_1: Higher schooling is associated with higher life expectancy.
- H_2: Developed countries have significantly higher life expectancy.
- H_3: Adult mortality has a strong negative effect.

4 Model Development

4.1 Model Selection

Justify regression and the candidate models (Linear, Ridge/Lasso, Random Forest, Gradient Boosting).

4.2 Feature Selection

Drop redundant pairs (e.g. `infant_deaths` vs. `under_five_deaths`); give the rule.

4.3 Splitting and Scaling

```
X_tr, X_te, y_tr, y_te = train_test_split(X, y, test_size=0.2, random_state=42)
scaler = StandardScaler().fit(X_tr[num_cols])      # fit on train only
```

Note on grouped data: consider splitting by country to avoid leakage across years.

4.4 Baseline and Training

Mean-prediction baseline RMSE, then each model's settings.

4.5 Hyperparameter Tuning

```
GridSearchCV(Ridge(), {"alpha":[0.01,0.1,1,10,100]}, cv=5,
             scoring="neg_root_mean_squared_error")
```

5 Results and Evaluation

5.1 Model Comparison

Table 6: Test-set performance (fill in).

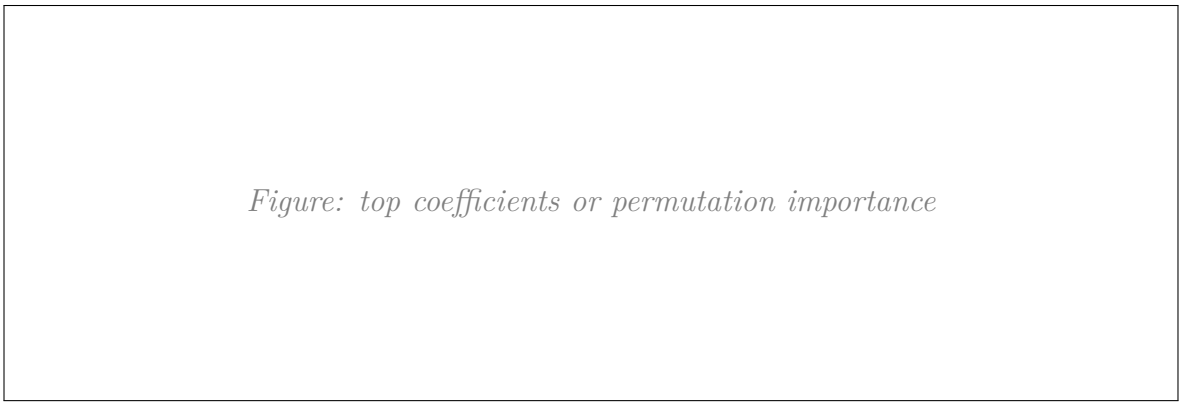
Model	Train RMSE	CV RMSE	Test RMSE	Test R^2
Baseline (mean)	XX	–	XX	0.00
Linear Regression	XX	XX	XX	XX
Ridge (tuned)	XX	XX	XX	XX
Random Forest	XX	XX	XX	XX
Gradient Boosting	XX	XX	XX	XX

Interpret: overfitting (train vs. test gap), best model and why.

5.2 Predicted vs. Actual and Residuals

Figure: predicted vs. actual scatter and residual plot

5.3 Feature Importance / Coefficients



Link back to the EDA hypotheses H_1–H_3: supported or not.

6 Conclusion and Business Implications

6.1 Key Findings

1. Finding 1 with a number.
2. Finding 2 with a number.

6.2 Implications

Who can use this (health ministries, NGOs, development agencies) and how.

6.3 Limitations

Imputed values, country-level aggregates (ecological fallacy), correlation $\neq$ causation, years pooled.

6.4 Future Work

Panel/time-series models, more recent data, interaction terms.

7 References

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8 Appendix

1. Source code link: `https://github.com/...`

2. Dataset link: `https://www.kaggle.com/datasets/kumarajarshi/life-expectancy-who`

3. Data dictionary (variable names after cleaning):

   - `country` (qualitative): country name
   - `year` (ordinal): year of observation, 2000–2015
   - `status` (qualitative): Developed or Developing
   - `life_expectancy` (quantitative, **target**): life expectancy at birth, in years
   - `adult_mortality` (quantitative): adult mortality rate, both sexes, per 1000 population
   - `infant_deaths` (quantitative): number of infant deaths per 1000 population
   - `alcohol` (quantitative): alcohol consumption per capita, in litres of pure alcohol
   - `percentage_expenditure` (quantitative): health expenditure as % of GDP per capita
   - `hepatitis_b` (quantitative): Hepatitis B immunization coverage, %
   - `measles` (quantitative): reported measles cases per 1000 population
   - `bmi` (quantitative): average body mass index of the population
   - `under_five_deaths` (quantitative): deaths under age five per 1000 population
   - `polio` (quantitative): polio immunization coverage, %

- `total_expenditure` (quantitative): government health expenditure as % of total government expenditure
- `diphtheria` (quantitative): diphtheria (DTP3) immunization coverage, %
- `hiv_aids` (quantitative): deaths per 1000 live births due to HIV/AIDS, ages 0–4
- `gdp` (quantitative): gross domestic product per capita, in USD
- `population` (quantitative): population of the country
- `thinness_1_19_years` (quantitative): prevalence of thinness among ages 10–19, %
- `thinness_5_9_years` (quantitative): prevalence of thinness among ages 5–9, %
- `income_composition` (quantitative): income composition of resources, index from 0 to 1
- `schooling` (quantitative): average years of schooling
- `region` (qualitative, engineered): region derived from country
- `period` (qualitative, engineered): year grouped into periods
- `income_tier` (qualitative, engineered): GDP group (low/medium/high)

4. Sample of the raw data: insert a screenshot of `data.head(10)`.
5. Supplementary figures and tables: extra plots, the full missing-value table, the full coefficient table, tuning results.